



Tug of war (also known as tug o' war, tug-a-war, tug war, rope war, rope pulling, or tugging war)



Team Project

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1. Project Definition and Team Formation	1
2. Background and Related Work	1
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4. Evaluation	
5. Demo: 3-Minute Talk (3MT)	
6. Project Management and Documentation (Grad Students Only)	

1. Proposal: Project Definition and Team Formation

Assignment

Due September 24 at 5:00 AM

Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

2. Background and Related Work

Assignment

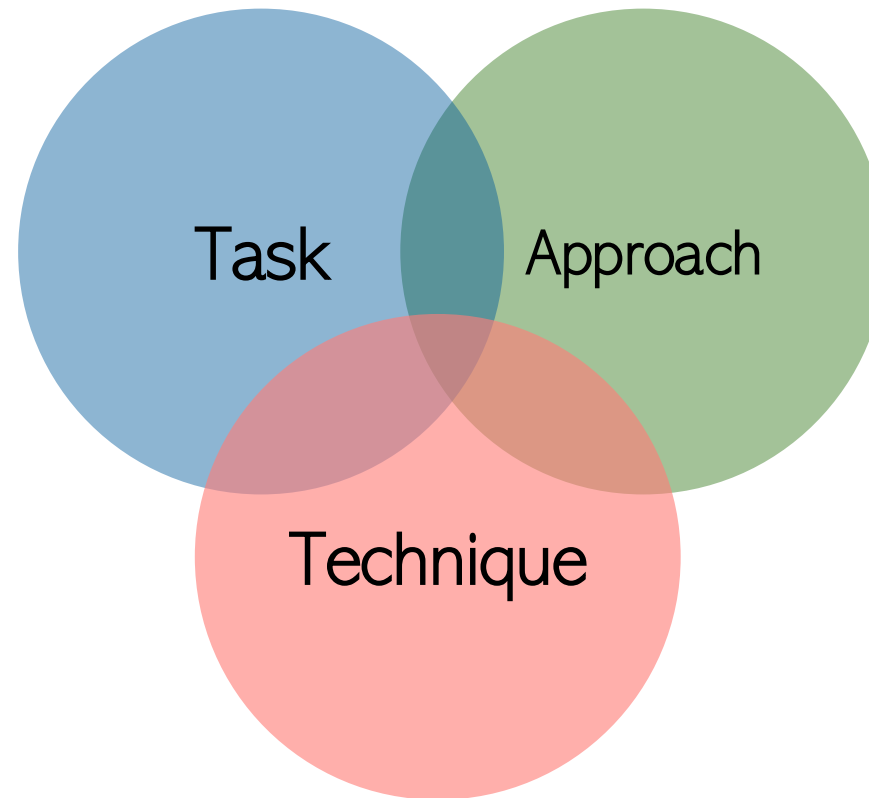
Due October 8 at 5:00 AM

Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

Outline: W02B: Topics Start

Classification

Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning

Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

4 Outline: W03A: Topics Start

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k -Nearest Neighbor

III) Evaluation

Aletheia or Alethia /æliˈθaɪ.ə/ is truth or disclosure in philosophy. Originating in Ancient Greek philosophy, used for the first time in by Parmenides in his poem On Nature. He contrasts it with doxa (opinion). In Greek mythology, Aletheia was personified as a Greek goddess of truth, sincerity and unconcealment.

"the daughter of Zeus, unforgetting Truth"



Supervision: The Way of Truth

Natural Language:

Learning to assign an observation to one of a predefined set of classes.

Who tells the assignment is true, correct, valid, ...?

Aletheia, the goddess of truth

God, in the Abrahamic religions as the all-powerful, all-knowing, eternal deity, and single creator of the universe

Omniscient /äm'niSHənt/: omni=all, scientia= knowledge or to know

The Oracle of Delphi

Teacher

Supervisor

Human (Expert) Annotator

Majority Vote on Human (Expert) Annotators

LLM-as-a-Judge



The oracle sat upon her tripod.
John Collier's The Priestess of Delphi, 1891

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Fully Supervised Classification

Natural Language:

Learning to assign an observation to ~~one~~ its **true class/label** in a predefined set of classes.

Formal Language:

Given a set of world observations and their true classes $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$ where $\mathbf{C}$ is a finite set, the task is to learn a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}$, such that for $(o, c) \in \mathbf{O}^*$, $f(o : \mathbf{O}^*) = c$.

Terminology

$\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$, labeled observations, experiences, or data set

Important:

f has **full access** to the entire $\mathbf{O}^*$



Fully Supervised Classification

We already have the labels $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$,

→ Why learning a function f to tell us an observation's label?

→ Preaching to the Converted

→ Preaching to the Choir /'kwɪər/

→ Summarize the $(o \leftrightarrow c)$ associations in $\mathbf{O}^*$

→ Denoising $(o \leftrightarrow c)$ in $\mathbf{O}^*$

Our focus!

→ Inferring/predicting unseen/unknown/missing $(\hat{o} \leftrightarrow c) \notin \mathbf{O}^*$

“An observation with no label/class: $(\hat{o} \leftrightarrow ?)$ ”

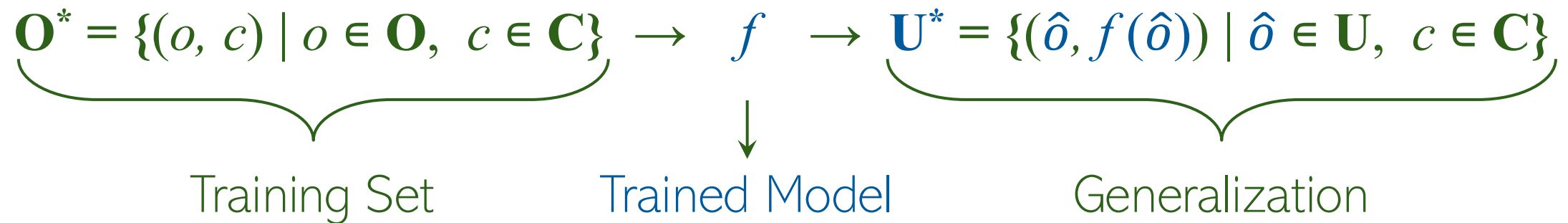
→ *Unseen to labeled dataset: $(\hat{o} \leftrightarrow c) \notin \mathbf{O}^*$*

→ *Unseen observation $(\hat{o})$ ”*

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Fully Supervised Classification: $\mathbf{O}^* \rightarrow \mathbf{U}^*$

Generalize to all observations of the universe $\mathbf{U}$, where $\mathbf{O}^*$ is a tiny subset of it!





Into the Wild, 2007, Sean Penn, Jon Krakauer
<https://www.youtube.com/watch?v=XZG1FzyB8DI>

Outline: W03A: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- III) Evaluation



Observations are pixels

Real-World Problem (Task): Coloring pixels (restore pixels' color)

Data Mining (Machine Learning) Task: Classification

Supervision: Observations' (pixels') classes (labels) are colors {Blue, Green, ...}

Unseen observation: Uncolored (damaged) pixel

Real-World Subproblem (Subtask): Restoration of the pixel's color

Data Mining (Machine Learning) Technique: k -Nearest Neighbor

Lin Chen

Ocean Photographer Of The Year 2026

https://www.nationalgeographic.pt/fotografia/imagens-que-mostram-beleza-oculta-mares-e-oceanos-concurso-ocean-photographer-of-the-year_7414

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k -Nearest Neighbor: Natural Language

"We become who we hang out with" - Napoleon Hill

"If you hang around 5 confident/intelligent/millionaires/idiots, you will be the 6th – Denzel Washington



Oliver Napoleon Hill (1883 –1970)

American self-help author and **conman**.

Known for his book *Think and Grow Rich* (1937), the best-selling self-help books of all time.

Most of his books are promoted as expounding principles to achieve "success".

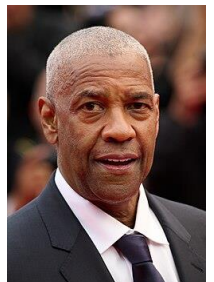
A controversial figure, **history of criminal charges and fraud accusations**, historians also doubt many of his claims.

Two Academy Awards, an Actor Award, two Golden Globes, two Silver Bears and a Tony Award as well as nominations for a Grammy Award and two Primetime Emmy Awards.

Denzel Hayes Washington Jr. (1954-Present)

American actor and producer.

Known for his dramatic roles on stage and screen.



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k -Nearest Neighbor: Math Language (Formal)

Given a set of world observations and their true classes $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$ where $\mathbf{C}$ is a finite set, k -Nearest Neighbor (k -NN) is a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}$, such that for an unseen observation $\hat{o} \in \mathbf{U}$,

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$

where

- $d: \mathbf{O} \times \mathbf{O} \rightarrow \mathbb{R}^{\geq 0}$ a function measuring distance between two observations;
- $k \in \mathbb{N}^+$ is the number of nearest neighbors considered;

k -Nearest Neighbor: Let's zoom in ...

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \text{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$


No internal memory (parameters)

No (pre)processing (training)

Only inference (prediction)

→ *non*-Parametric (no summarization)

→ Lazy learner (no learning at all)

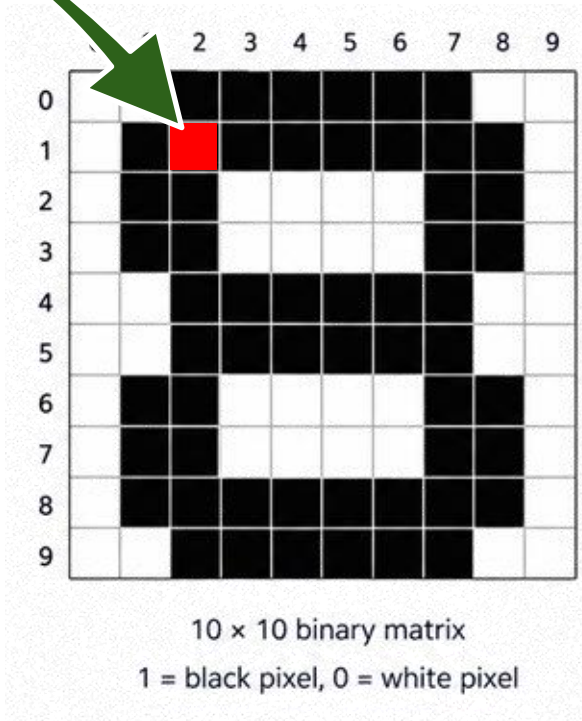
→ Instance-based learner

Missing value inference

Denoising

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k -Nearest Neighbor: kNN.ipynb



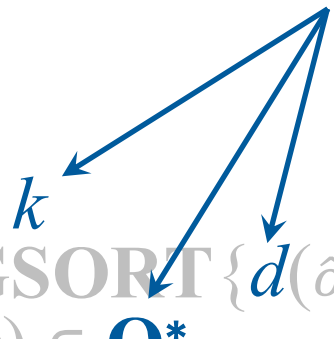
Outline: W03A: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
Let's zoom in ... Hyperparameters
- III) Evaluation



Tug of war (also known as tug o' war, tug-a-war, tug war, rope war, rope pulling, or tugging war)

k -Nearest Neighbor: Let's zoom in ...

$$f(\hat{o} : \mathbf{O}^*, k, d) = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \text{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \mathbf{1} [c_i = c] \right\}$$


No internal memory (parameters) $\rightarrow$ *non*-Parametric (no summarization)

But it has Hyperparameters

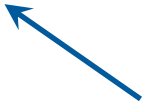
$\rightarrow$ Compile-time parameters

Form the learning algorithm architecture (structure)

Are “*tuned*” during design to find the best architecture

Remain constant after deploy for production

k -Nearest Neighbor: Efficacy vs. Efficiency Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \text{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


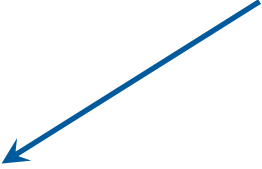
Scaling labeled dataset:

- Small → Fast but Poor
- Large → Slow but Accurate

Labeled set of observations (training set) is commonly not considered a hyperparameter. Why?

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k -Nearest Neighbor: Variance vs. Bias Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


k , number of nearest neighbor

→ Small like $k = 1$

Liberal: decides based on whatever sits next to $\hat{o}$

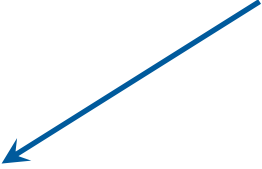
Unbiased (Low Bias): not impose a strong global preference

Locally Sensitive

High Variance

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k -Nearest Neighbor: Variance vs. Bias Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


k , number of nearest neighbor

→ Large like $k = |\mathbf{O}^*|$

Conservative: considers a broader neighborhood

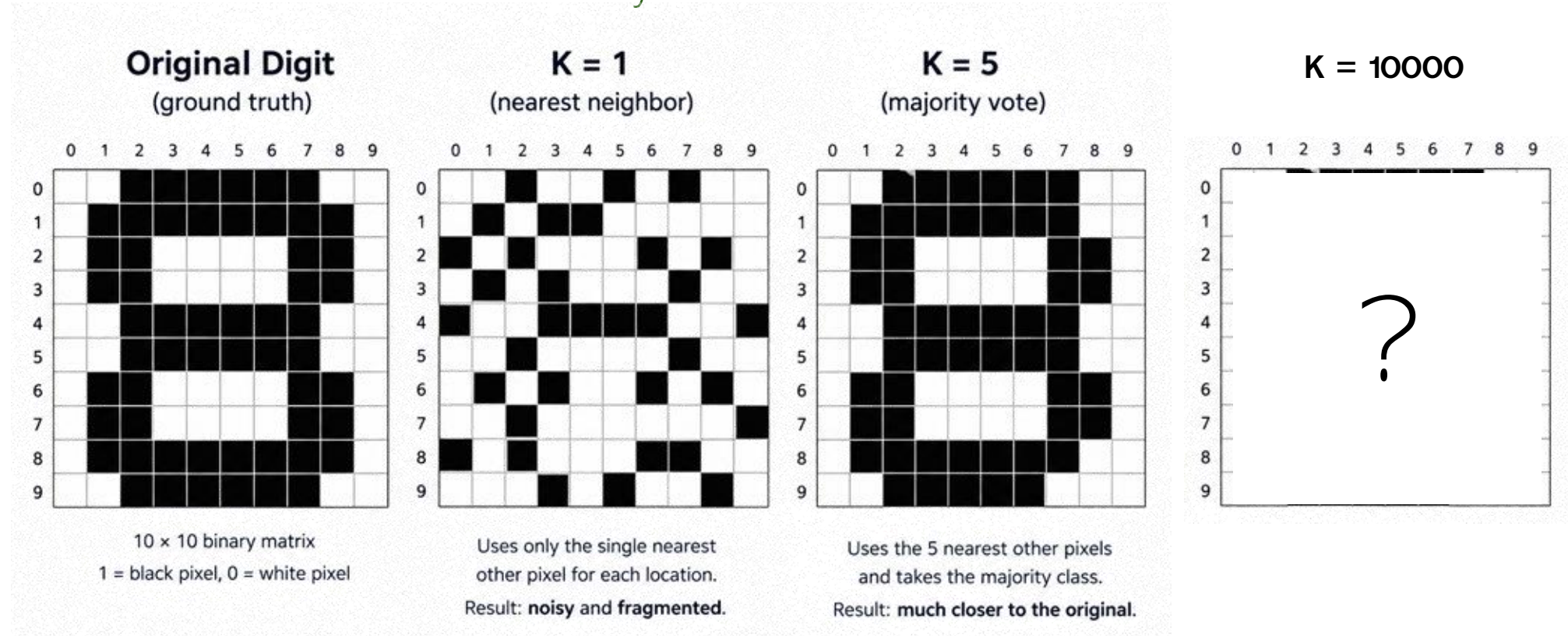
Low Variance

Biased (High Bias): Blind, favors the global over local pattern

Globally Conforming, Another Brick in the Wall :|

k -Nearest Neighbor: Inference on Training Set

A good learner (technique) can easily learn the training set.
A student whose exam contains only the lecture contents.



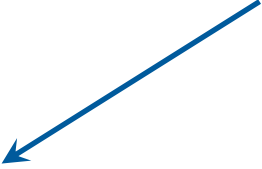
We make a copy of an image, then restore every pixel of it by its surrounding ...



Daniel Edmond Sullivan Jr
Ocean Photographer of the Year
<https://www.nationalgeographic.pt/fotografia/imagens-que-mostram-beleza-oculta-mares-e-oceanos-concurso-ocean-photographer-of-the-year> 7414

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 r -Nearest Neighbor: Neighbors vs. Neighborhood

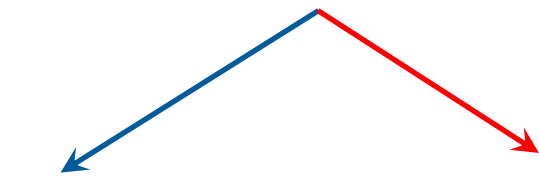
$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \underset{d \leq r}{1} [c_i = c] \right\}$$


$k \in \mathbb{N}^+$ number of nearest neighbors $\rightarrow r \in \mathbb{R}^{\geq 0}$ radius of nearest neighborhood

- $\rightarrow$ Small, like r close to 0 $\rightarrow$ Empty neighborhood, cannot decide.
- $\rightarrow$ Large, like $r = +\infty \rightarrow$ Very crowded neighborhood, conformity.

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k -Nearest Neighbor: Weighted (“Attentive”)



$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \frac{1}{k} [c_i = c] \right\}$$

k , number of nearest neighbor

→ Within the top- k , all observations are treated equally.

Equality is not fair; equity is.

→ Farther observations deserve less attention.

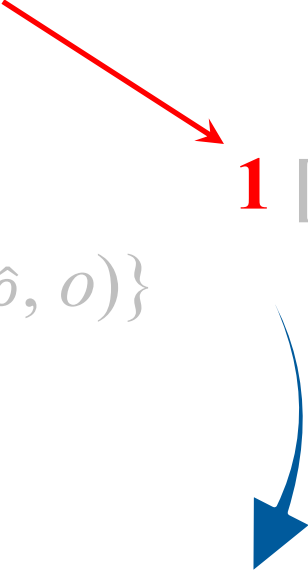
→ E.g., should I go to university or college?

Parents' labels counts as normal (1x)

Friends' labels counts half (0.5x)

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 k -Nearest Neighbor: Weighted (“Attentive”)

$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX}_{c_i \in \mathbf{C}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$


$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX}_{c_i \in \mathbf{C}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \frac{1}{d(\hat{o}, o)} \times [c_i = c] \right\}$$


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 k -Nearest Neighbor: Distance Function

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$
